

Defining Semantic Structure: Model Theory

A Detailed Explanatory Chapter

Introduction

Formal semantics aims to define meaning in precise mathematical terms. To do so, we must answer a fundamental question: what does it mean for a sentence to be true? Model theory provides the formal apparatus for answering this question. Rather than defining meaning psychologically or intuitively, model theory defines truth relative to mathematical structures called models. A model represents a possible world in simplified, abstract form. Sentences are interpreted relative to such models, and their truth conditions are defined recursively according to their logical structure.

Model theory therefore provides the foundation of formal semantics. It explains how predicates denote sets, how quantifiers range over domains, how logical connectives determine truth conditions, and how logical consequence is defined. This chapter develops a detailed understanding of how semantic structure is defined using model theory.

1. The Idea of a Model

A model is a mathematical structure that interprets the symbols of a formal language. In natural language semantics, a model corresponds to a possible world or situation. It specifies which individuals exist and which properties and relations hold among them.

Formally, a model $\mathcal{M}$ is defined as:

$$\mathcal{M} = \langle D, I \rangle$$

where:

- D is a non-empty set called the domain.

- I is an interpretation function.

The domain D contains the entities under discussion. For example, if we are discussing students and books, the domain may contain all individuals relevant to that discourse.

The interpretation function assigns meanings to non-logical symbols:

- Each constant symbol is assigned an element of D .
- Each unary predicate is assigned a subset of D .
- Each binary predicate is assigned a subset of $D \times D$.

Thus, a predicate denotes a set (or relation), and a constant denotes an individual.

2. Interpretation of Terms

Terms denote individuals. Their interpretation depends on the model.

If a is a constant, then:

$$a^{\mathcal{M}} = I(a)$$

This means the denotation of a is the object assigned by the interpretation function.

Variables require an assignment function g :

$$g : Var \rightarrow D$$

If x is a variable, then:

$$x^{\mathcal{M},g} = g(x)$$

Variables do not have fixed denotations; they receive values from the assignment.

3. Interpretation of Atomic Formulas

Consider an atomic formula:

$$P(a)$$

This is true in model $\mathcal{M}$ if and only if:

$$a^{\mathcal{M}} \in I(P)$$

In words, the individual denoted by a must belong to the set denoted by P .

For a binary predicate:

$$R(a, b)$$

This is true if:

$$\langle a, b \rangle \in I(R)$$

Thus, atomic truth is defined via set membership.

4. Recursive Definition of Truth

Truth is defined recursively based on logical structure.

Negation

$$\mathcal{M} \models \neg\phi$$

iff ϕ is not true in $\mathcal{M}$.

Conjunction

$$\mathcal{M} \models \phi \wedge \psi$$

iff both ϕ and ψ are true in $\mathcal{M}$.

Universal Quantification

$$\mathcal{M} \models \forall x \phi$$

iff for every $d \in D$:

$$\mathcal{M} \models \phi[x \mapsto d]$$

This means that ϕ must hold for every object in the domain.

Existential Quantification

$$\mathcal{M} \models \exists x \phi$$

iff there exists some $d \in D$ such that:

$$\mathcal{M} \models \phi[x \mapsto d]$$

Quantifiers therefore range over domain elements.

5. Step-by-Step Evaluation Example

Consider:

$$\forall x (Student(x) \rightarrow Passed(x))$$

Let the model be:

$$D = \{a, b\}$$

$$I(Student) = \{a, b\}$$

$$I(Passed) = \{a\}$$

Evaluation:

- For $x = a$: Student(a) is true; Passed(a) is true.
- For $x = b$: Student(b) is true; Passed(b) is false.

Thus the implication fails for b , so the universal statement is false in this model.

This example shows how truth is computed by systematic checking over domain elements.

6. Truth in a Model vs Logical Validity

A sentence is true in a model if it holds in that specific structure.

A sentence is logically valid if it is true in all models.

For example:

$$\forall x P(x) \rightarrow \exists x P(x)$$

is valid in every non-empty domain.

Logical validity is defined as:

$$\models \phi$$

meaning ϕ is true in all models.

7. Logical Consequence

A conclusion ψ follows from premises Γ if:

$$\Gamma \models \psi$$

This means every model that makes all formulas in Γ true also makes ψ true.

Thus inference is defined semantically.

8. Satisfiability and Contradiction

A formula is satisfiable if there exists some model in which it is true.

It is contradictory if it is false in all models.

Example:

$$P(a) \wedge \neg P(a)$$

is false in every model and thus contradictory.

9. Isomorphism and Structural Invariance

Two models are isomorphic if there is a one-to-one mapping between their domains preserving relations.

Isomorphic models satisfy the same sentences. This shows that logic captures structural relationships rather than accidental naming differences.

10. Application to Natural Language

Natural language semantics uses model theory to define truth conditions.

Sentence:

Rina loves Amit.

Logical form:

$$\text{Loves}(r, a)$$

This is true in model $\mathcal{M}$ if:

$$\langle I(r), I(a) \rangle \in I(Loves)$$

Thus meaning is defined as truth conditions relative to a model.

Quantified sentence:

Every student laughed.

Logical form:

$$\forall x(Student(x) \rightarrow Laugh(x))$$

This is true in a model if the set denoted by Student is a subset of the set denoted by Laugh.

Conclusion

Model theory provides the mathematical foundation for defining semantic structure. By specifying domains, interpretation functions, and recursive truth conditions, it defines precisely what it means for a sentence to be true. Logical consequence, validity, and satisfiability are defined in terms of truth across models. This framework allows natural language meaning to be analyzed with mathematical rigor and clarity. Model theory thus forms the backbone of formal semantics.